Appendix 1 Stepwise Logistic regression models of EGG parameter measures for individual languages: CQ, closing duration, PIC and PDC

Bo

	Df	Deviance	AIC	LRT	*p*-value(Chi)	
<none>		1641.2	1643.2			
CQ	1	1482.8	1486.8	158.476	**2.20E-16**	***
SQ	1	1640.5	1644.5	0.789	0.37451	
Opening	1	1640.6	1644.6	0.63	0.42718	
Closing	1	1612.4	1616.4	28.81	**7.99E-08**	***
PIC	1	1618.2	1622.2	23.071	**1.56E-06**	***
PDC	1	1543.1	1547.1	98.146	**2.20E-16**	***
PIC_time	1	1637.6	1641.6	3.642	0.05633	.
PDC_time	1	1637.7	1641.7	3.533	0.06017	.

Yi

	Df	Deviance	AIC	LRT	*p*-value(Chi)	
<none>		1463.8	1465.8			
CQ	1	1337.6	1341.6	126.199	**2.20E-16**	***
SQ	1	1462.8	1466.8	1.023	0.3118631	
Opening	1	1462.9	1466.9	0.943	0.3315606	
Closing	1	1463	1467	0.787	0.3749866	
PIC	1	1450.4	1454.4	13.455	**0.0002444**	***

PDC	1	1402.4	1406.4	61.407	**4.64E-15**	***
PIC_time	1	1458.8	1462.8	5.072	0.0243112	*
PDC_time	1	1458.9	1462.9	4.888	0.0270397	*

Hani

	Df	Deviance	AIC	LRT	*p*-value(Chi)	
<none>		2255.4	2257.4			
CQ	1	1954.7	1958.7	300.712	**2.20E-16**	***
SQ	1	2082.7	2086.7	172.702	**2.20E-16**	***
Opening	1	2238.4	2242.4	17.062	3.62E-05	***
Closing	1	2163.1	2167.1	92.366	**2.20E-16**	***
PIC	1	2188.2	2192.2	67.26	**2.38E-16**	***
PDC	1	2100.7	2104.7	154.694	**2.20E-16**	***
PIC_time	1	2224.8	2228.8	30.64	3.11E-08	***
PDC_time	1	2225.7	2229.7	29.741	4.94E-08	***

Appendix 2: Relation of EGG pulse shape to vocal fold vibration

Comparisons of EGG signals with more direct physiological signals (e.g. imaging techniques) have been made in order to understand the important features of the pulses and what they correspond to (e.g. Rothenberg *et al*., 1988; Childers *et al*., 1990; among many others). Researchers (Titze 1984, 1989, 1990; Childers *et al*., 1986; Larson *et al.*, 1994; among many others) have found that changes in certain basic geometric properties of EGG pulses reflect specifiable changes in vocal fold adjustment and behavior, especially the contact patterns (see Baken and Orlikoff 2000: 422 for detailed review). The right panel of Figure 1 below is a summary interpretation from various such comparison studies (see Baken and Orlikoff 2000 for review). Previous efforts to relate EGG pulse shapes to this kind of schematic (e.g. Marasek, 1997) have been difficult due to reliance on signal landmarks, and pulse variability across individual speakers, especially by gender. Consideration of EGG pulse shape components from FDA provides a different approach to such modeling. The left panel of the figure below reproduces part of our Figure 5 the first principal component from our FDA analysis, which is strongly related to tense vs. lax phonation.

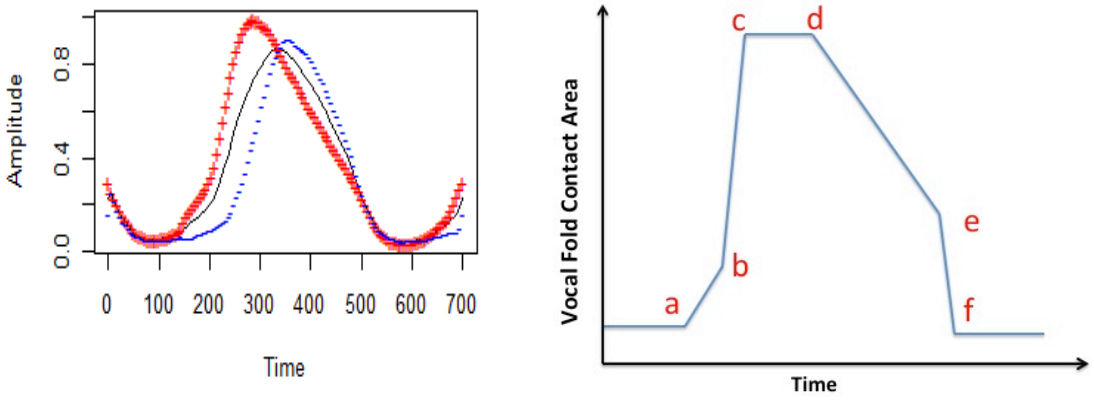


FIG. 1. Left panel is a single pulse of PC1; right panel are the annotations of critical points of glottal events. a=lower margins contact; b= upper margins contact; c-d=maximal contact; d= upper margins de-contact; e=lower margins de-contact; f=minimal contact.

In the right panel, point a is where the lower margins of the vocal folds make initial contact. From points a to b, the approximation of the lower margins of the folds speeds up. If the folds remain virtually parallel and closure occurs very rapidly along their length, then the slope of a-b is great. However, if the vocal folds have a gradual "zippering" closure, meaning that the folds form an acute angle (giving the glottis a V-shape), then the slope is less. Compare this part of the schematic with the corresponding part of the pulse in the left panel. We can see that the tense phonation (the red positive line) has a greater a-b slope, indicating a more abrupt contact of the lower margins. By contrast, the lax phonation (the blue negative line) has a much flatter slope. Moreover, it takes longer for the lax phonation to reach the initial change in contact (point a).

At Point b, the upper margins of the vocal folds make initial contact. During b-c, upper margin approximation spreads antero-posteriorly, and glottal closure is achieved during this interval. As for the a-b interval, a more abrupt contact would cause a steeper slope of

b-c. Compare this part of the schematic with the corresponding part of the pulse in the left panel. It can be seen that the tense phonation has a steeper b-c interval, suggesting that the contacting is more abrupt. By contrast, the point b for the lax phonation is lower and later, and the b-c interval has a shallower slope. Moreover, the maximum contact that the tense phonation can reach is greater than the lax phonation, suggesting a more complete or tight contact.

The interval from c to d is the phase when contact remains maximal, and then at point d, separation of the lower margins of the vocal folds begins, initiating the opening phase. Point e is when the separation of the lower margins is complete, and the upper margins begin to separate, resulting in a greater increase in the impedance. A knee at this point is arguably caused by the sudden rupturing of a bridge of mucus linking the vocal folds. Compare this part of the schematic with the corresponding part of the pulse in the left panel. Overall, the tense phonation has a clearer knee than the lax phonation, indicating a more abrupt change in contact. The de-contacting is at first slower in the tense phonation than in the lax phonation, but accelerates when the upper margins begin to separate (point e).

During the interval from e to f, the upper margins of the vocal folds continue to separate. Glottal opening occurs somewhere in the interval, following which glottal length and glottal width increase. However, the exact time within this interval at which a glottal space first appears is unknowable. Point f is where contact between the vocal folds reaches its minimum. From point f to point a is where the vocal folds are open with

minimum contact. Compare this part of the schematic with the corresponding part of the pulse in the left panel. The f-a interval is shorter for the tense phonation, compared to the lax phonation.

In sum, the tense phonation has a more abrupt and more complete contact, but a shorter duration and (perhaps) smaller extent of opening. As a result of the abrupt contact, the slope of the closing phase is steeper for the tense phonation. By contrast, the lax phonation has more gradual and less complete contact, and greater opening. The gesture must be very smooth, as the pulse is very symmetric and sine-like.